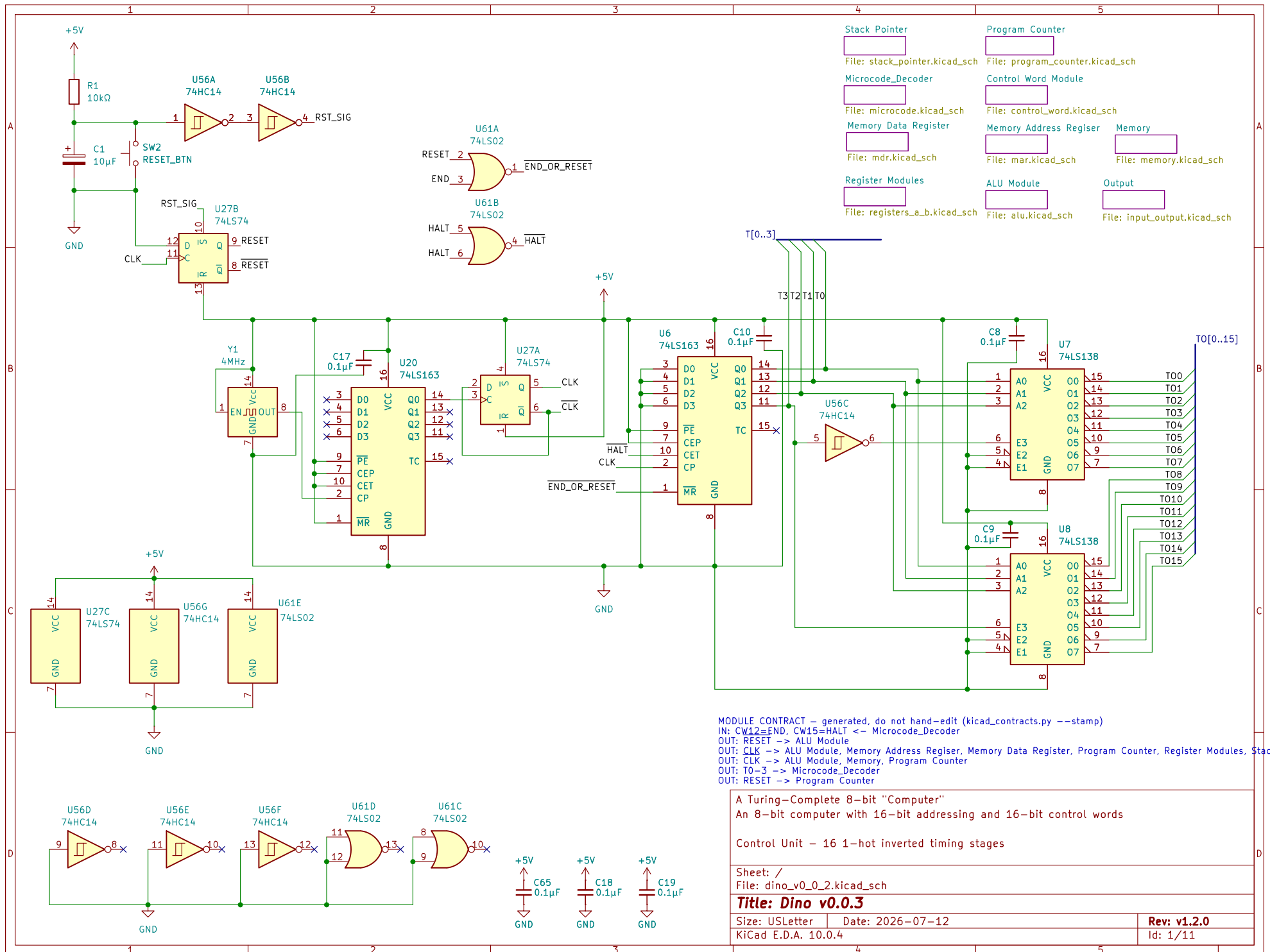
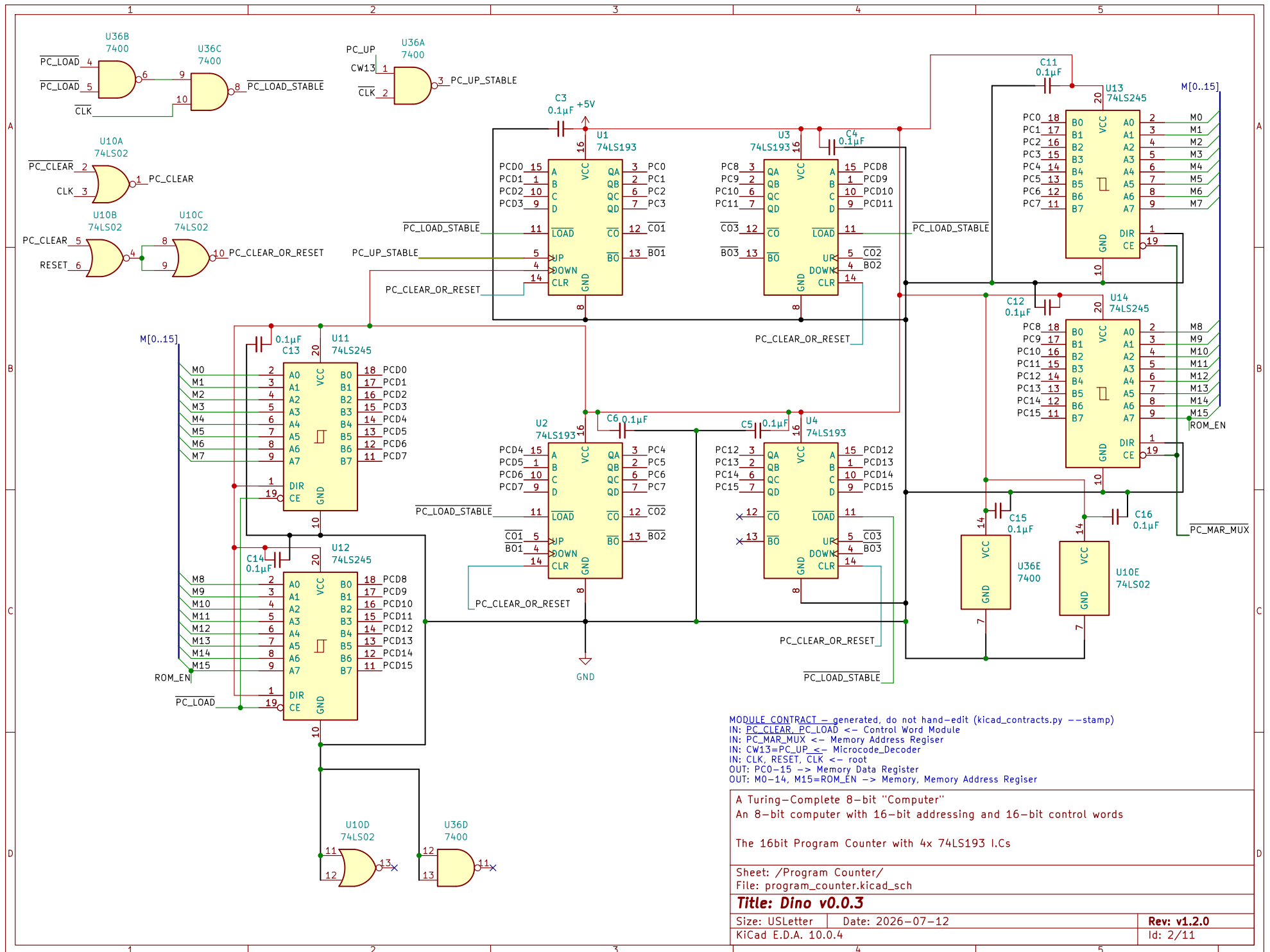
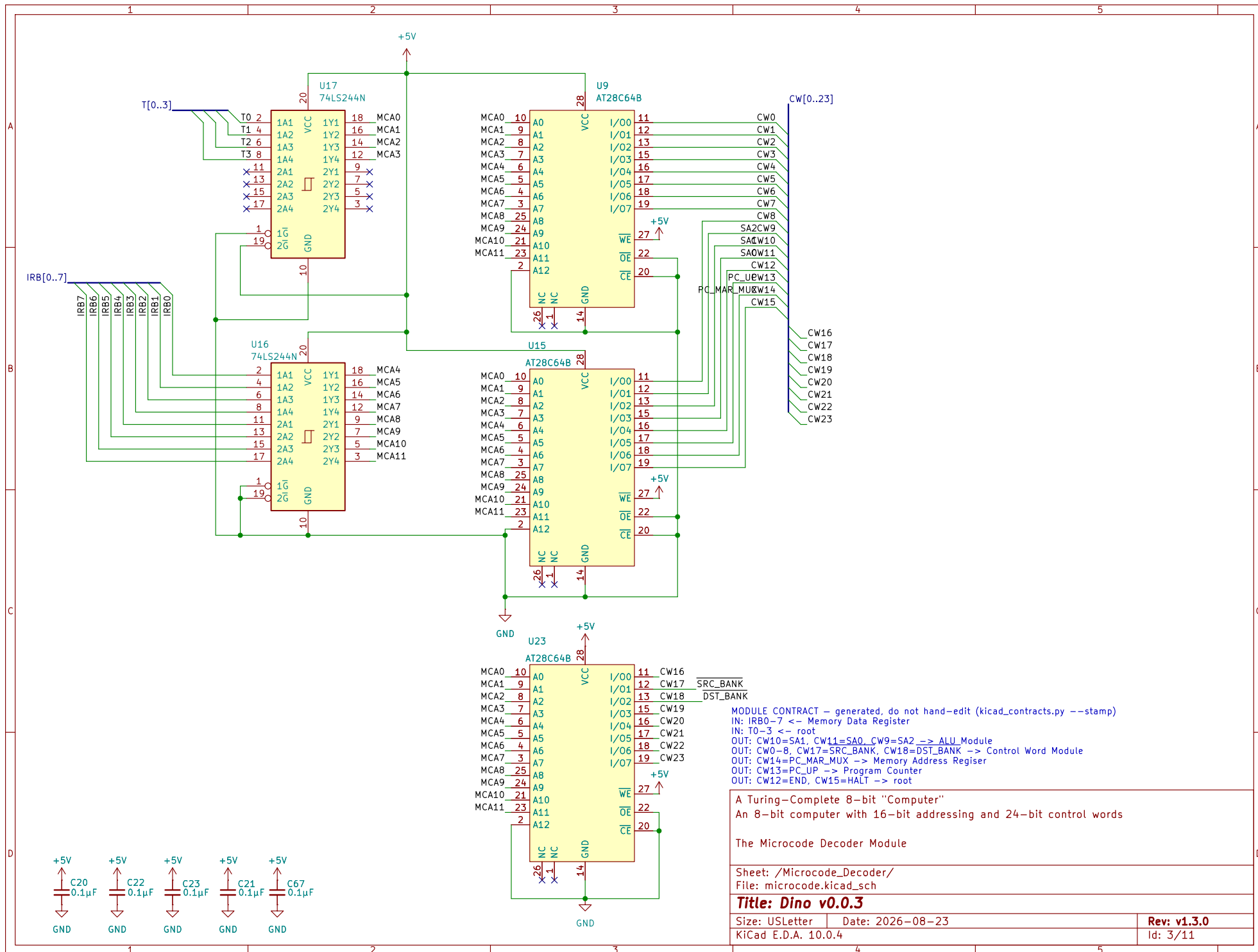


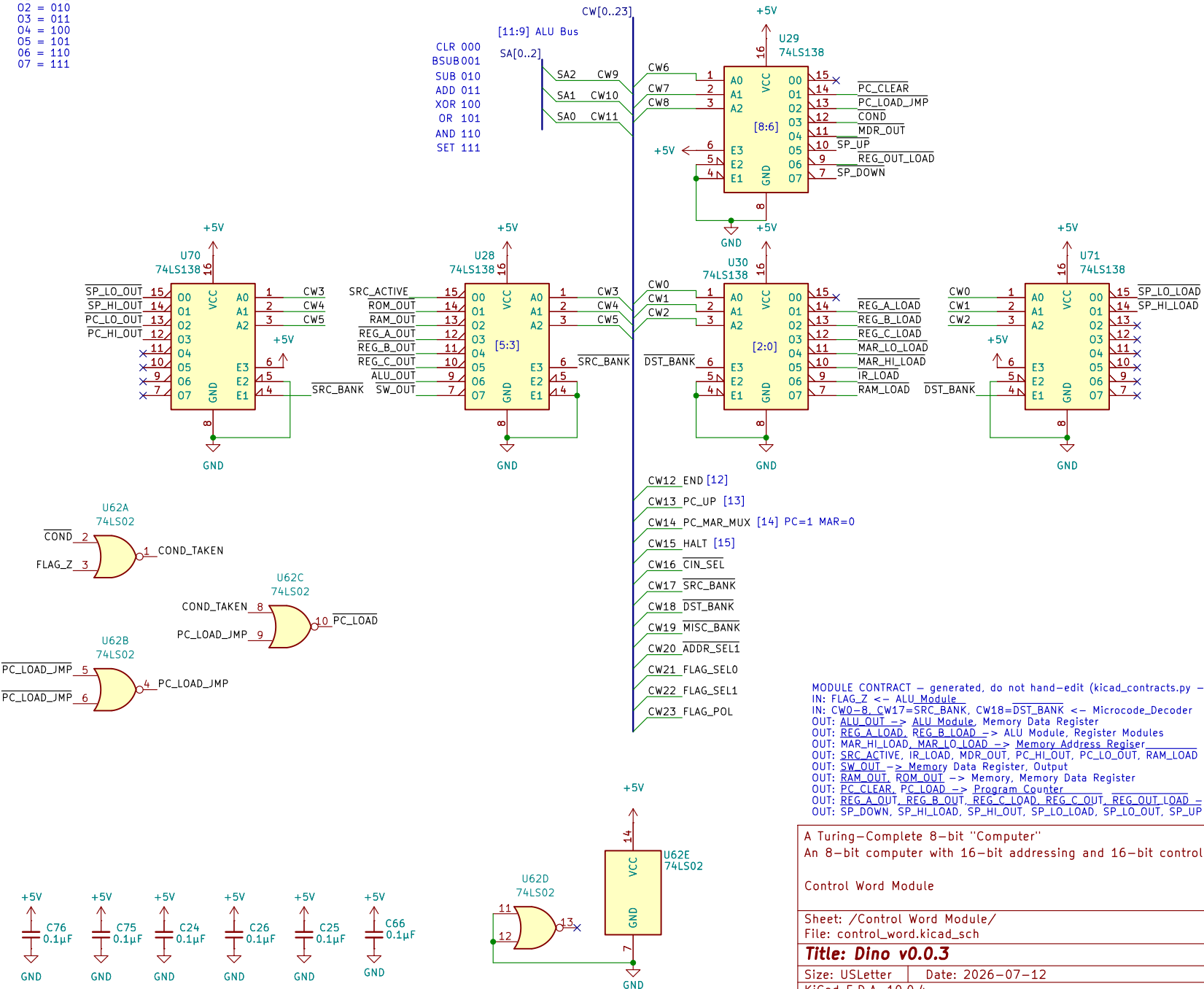






'138 Decoding  
 00 = 000  
 01 = 001  
 02 = 010  
 03 = 011  
 04 = 100  
 05 = 101  
 06 = 110  
 07 = 111

CLR 000  
 BSUB 001  
 SUB 010  
 ADD 011  
 XOR 100  
 OR 101  
 AND 110  
 SET 111



MODULE CONTRACT — generated, do not hand-edit (kicad\_contracts.py --stamp)  
 IN: FLAG\_Z <- ALU\_Module  
 IN: CW0-8, CW17=SRC\_BANK, CW18=DST\_BANK <- Microcode\_Decoder  
 OUT: ALU\_OUT -> ALU\_Module, Memory Data Register  
 OUT: REG\_A\_LOAD, REG\_B\_LOAD -> ALU\_Module, Register Modules  
 OUT: MAR\_HI\_LOAD, MAR\_LO\_LOAD -> Memory Address Register  
 OUT: SRC\_ACTIVE, IR\_LOAD, MDR\_OUT, PC\_HI\_OUT, PC\_LO\_OUT, RAM\_LOAD -> Memory Data Register  
 OUT: SW\_OUT -> Memory Data Register, Output  
 OUT: RAM\_OUT, ROM\_OUT -> Memory, Memory Data Register  
 OUT: PC\_CLEAR, PC\_LOAD -> Program Counter  
 OUT: REG\_A\_OUT, REG\_B\_OUT, REG\_C\_LOAD, REG\_C\_OUT, REG\_OUT\_LOAD -> Register Modules  
 OUT: SP\_DOWN, SP\_HI\_LOAD, SP\_HI\_OUT, SP\_LO\_LOAD, SP\_LO\_OUT, SP\_UP -> Stack Pointer

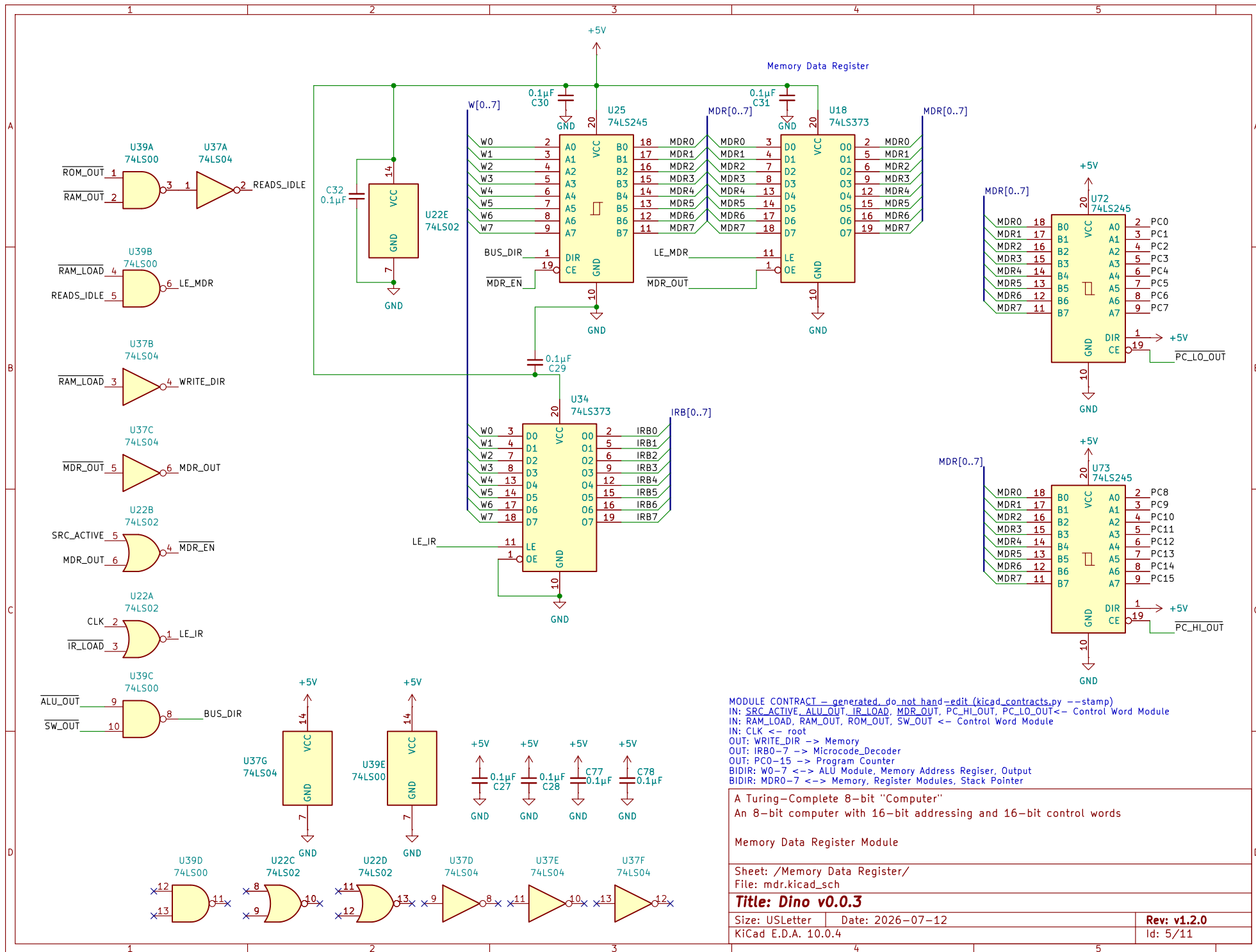
A Turing-Complete 8-bit "Computer"  
 An 8-bit computer with 16-bit addressing and 16-bit control words

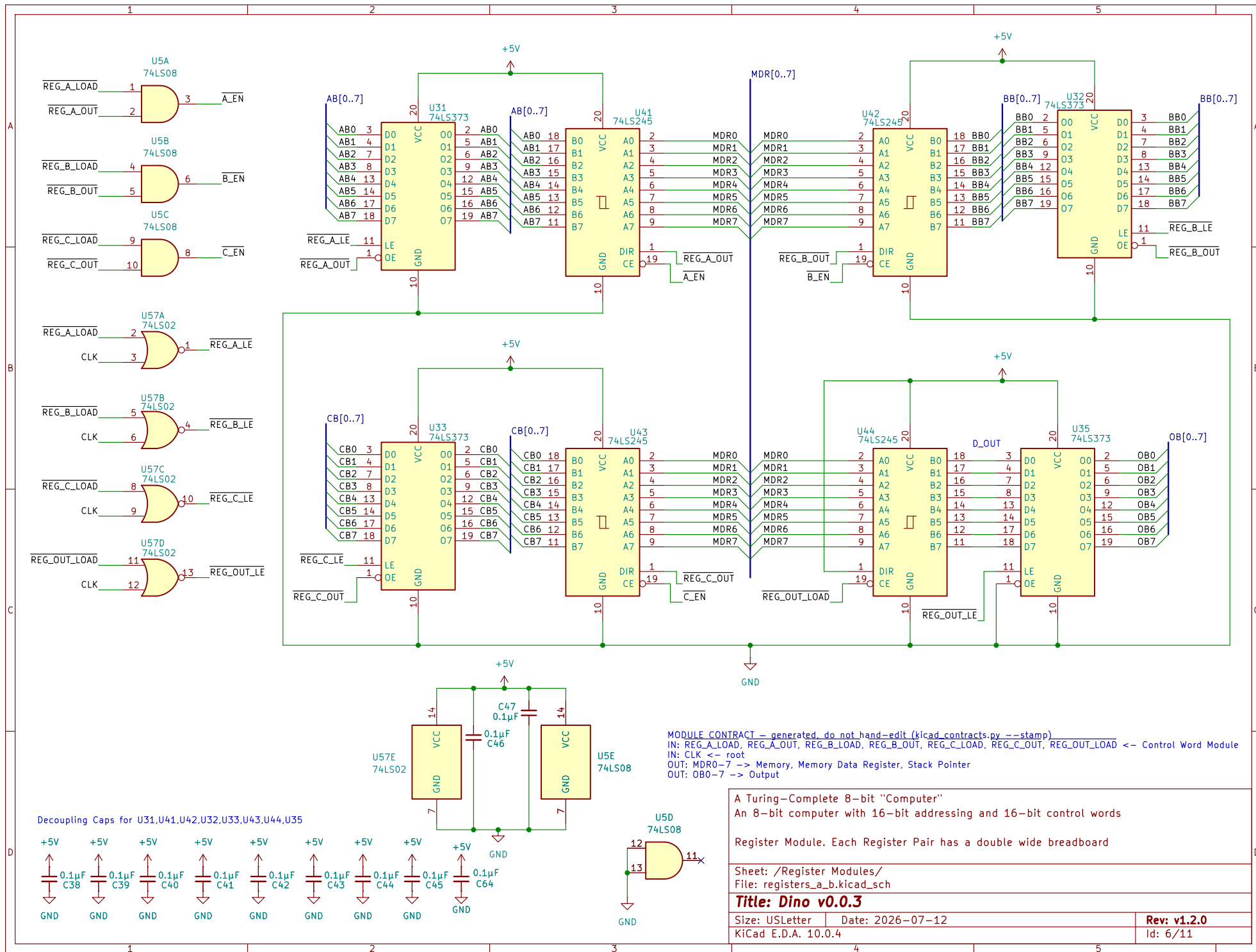
Control Word Module

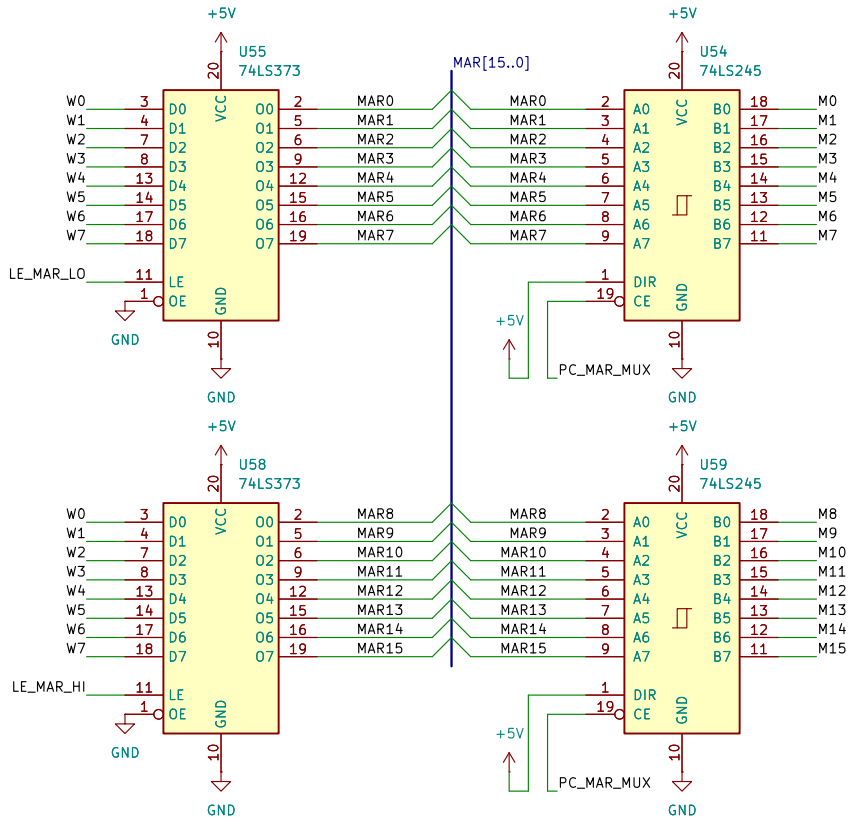
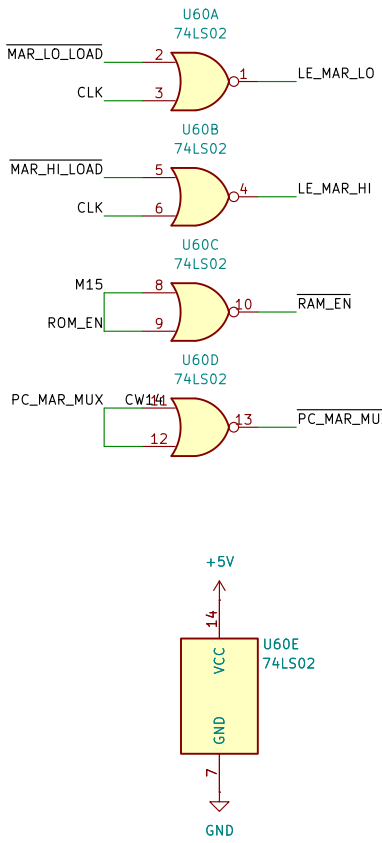
Sheet: /Control Word Module/  
 File: control\_word.kicad\_sch

**Title: Dino v0.0.3**

|                     |                  |             |
|---------------------|------------------|-------------|
| Size: USLetter      | Date: 2026-07-12 | Rev: v1.2.0 |
| KiCad E.D.A. 10.0.4 |                  | Id: 4/11    |



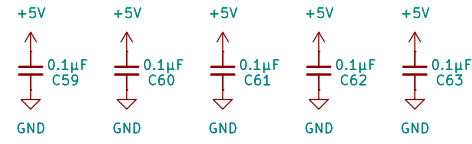


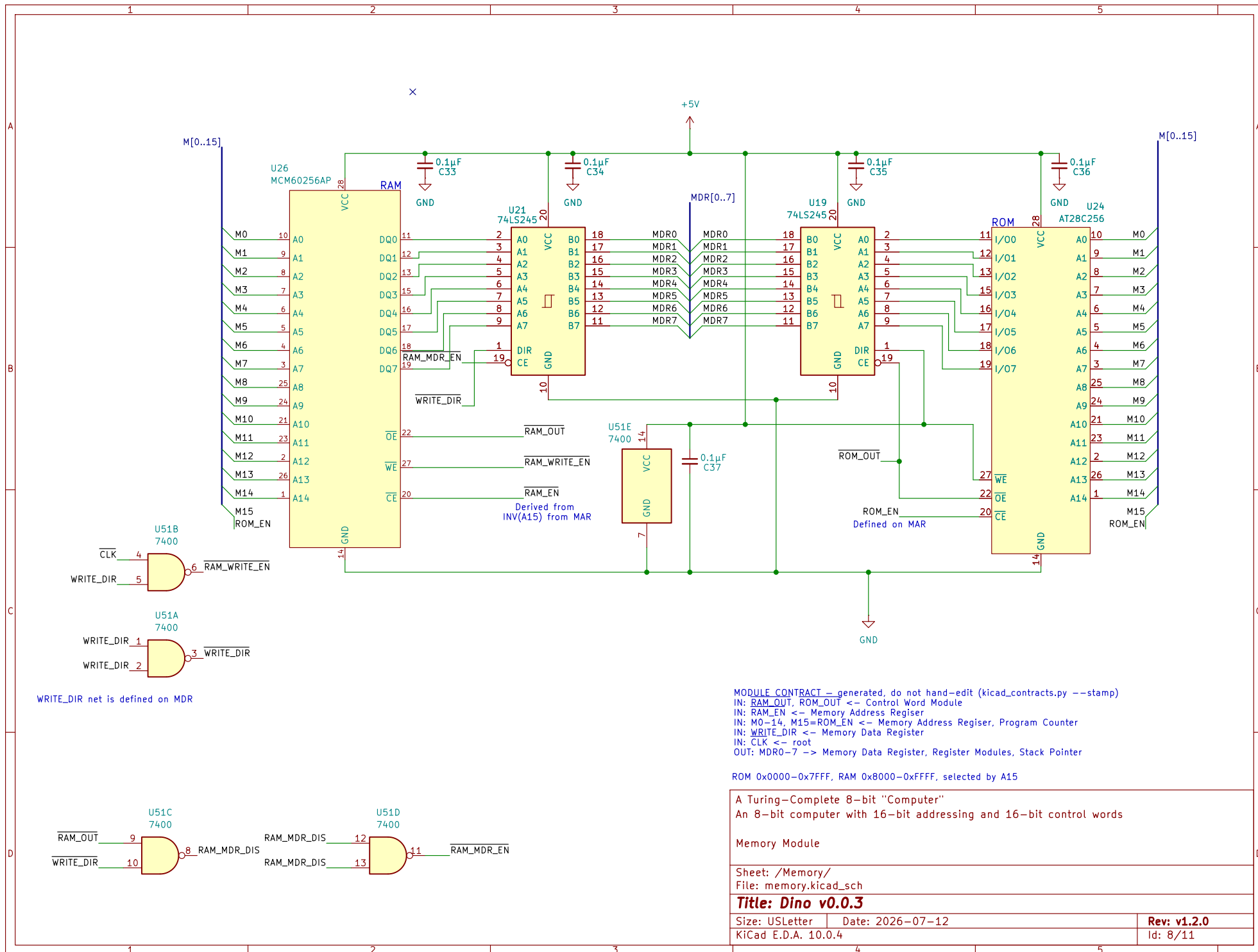


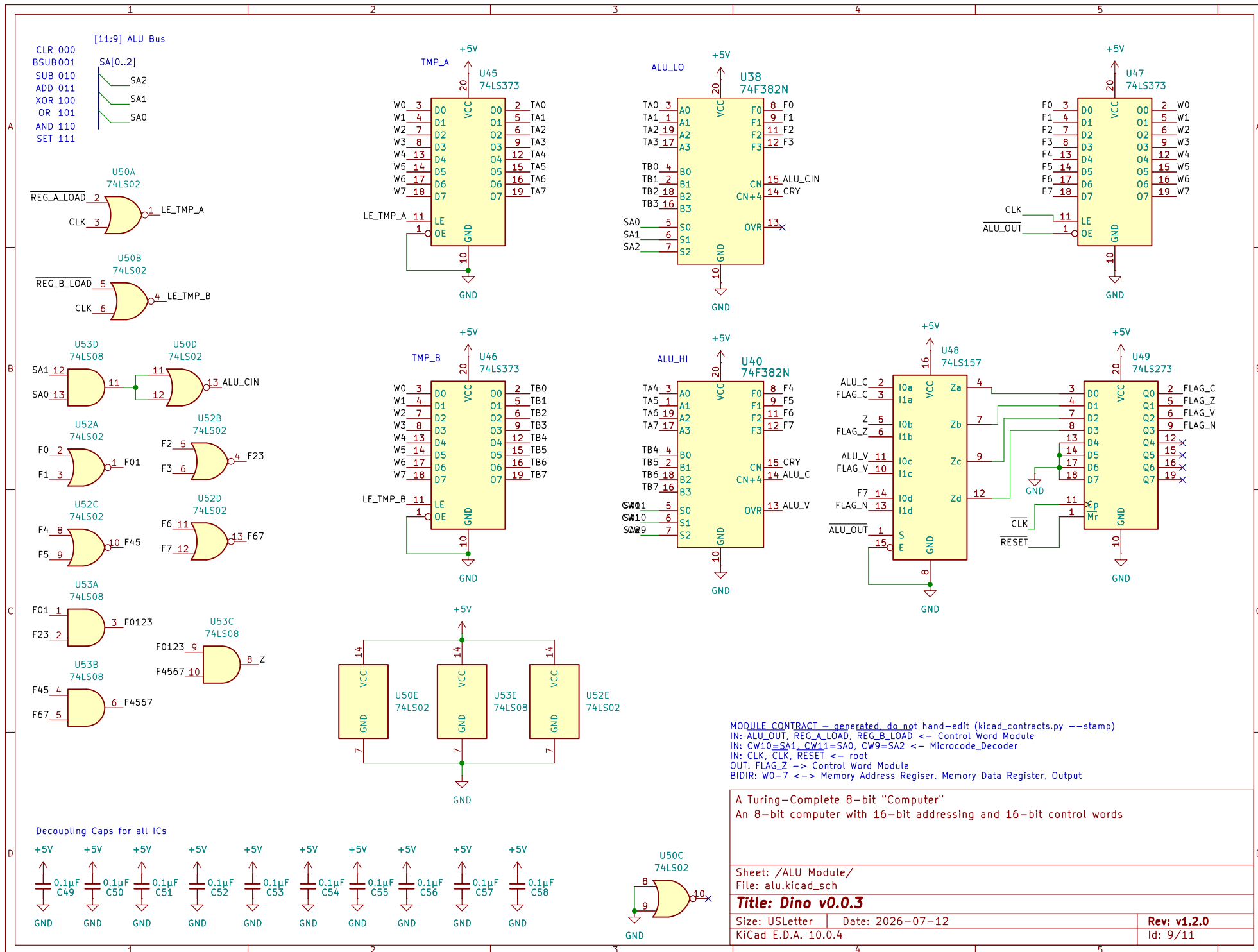
MODULE CONTRACT -- generated, do not hand-edit (kicad\_contracts.py --stamp)  
IN: W0-7 <= ALU\_Module\_Memory Data Register, Output  
IN: MAR\_HI\_LOAD, MAR\_LO\_LOAD <= Control Word Module  
IN: CW14=PC\_MAR\_MUX <= Microcode\_Decoder  
IN: CLK <= \_root  
OUT: RAM\_EN -> Memory  
OUT: M0-14 -> Memory, Program Counter  
OUT: PC\_MAR\_MUX -> Program Counter  
BIDIR: M15=ROM\_EN <=> Memory, Program Counter

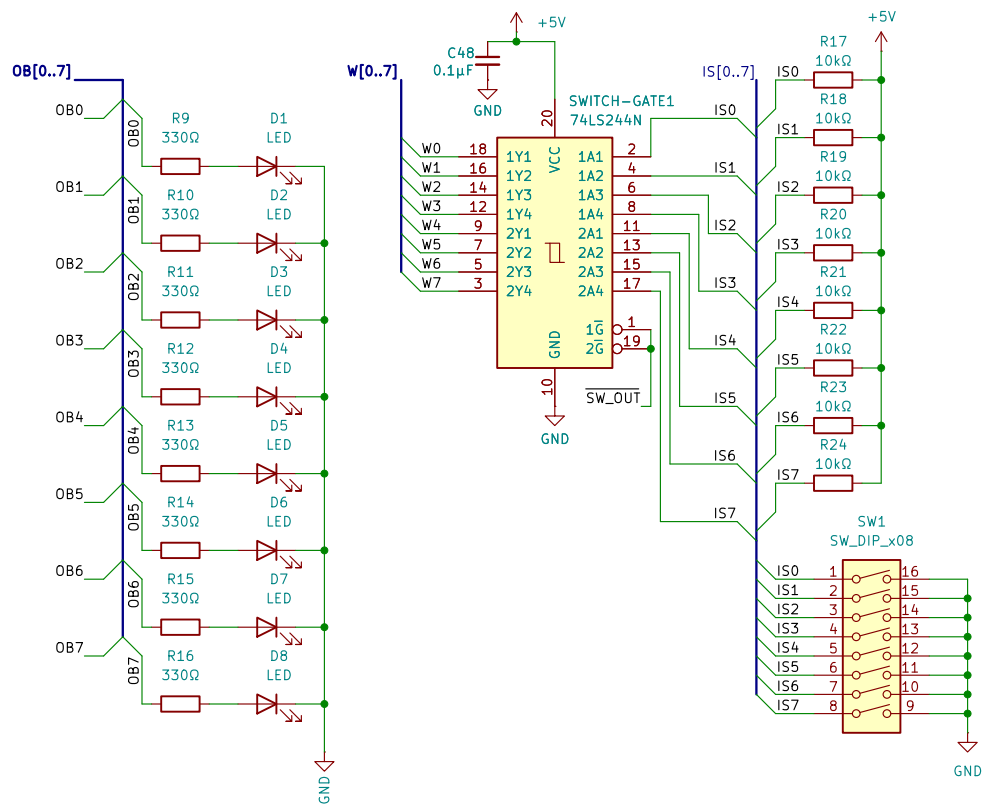
A Turing-Complete 8-bit "Computer"  
An 8-bit computer with 16-bit addressing and 16-bit control words

|                                                         |                         |
|---------------------------------------------------------|-------------------------|
| Sheet: /Memory Address Register/<br>File: mar.kicad_sch |                         |
| <b>Title: Dino v0.0.3</b>                               |                         |
| Size: A4                                                | Date: 2026-07-12        |
| KiCad E.D.A. 10.0.4                                     | Rev: v1.2.0<br>Id: 7/11 |









MODULE CONTRACT – generated, do not hand-edit (kicad\_contracts.py --stamp)  
 IN: SW\_OUT <- Control Word Module  
 IN: OB0-7 <- Register Modules  
 OUT: W0-7 -> ALU Module, Memory Address Register, Memory Data Register

A Turing-Complete 8-bit "Computer"  
 An 8-bit computer with 16-bit addressing and 16-bit control words

Input Output Components

Sheet: /Output/  
 File: input\_output.kicad\_sch

**Title: Dīno v0.0.3**

Size: USLetter Date: 2026-07-12  
 KiCad E.D.A. 10.0.4

Rev: v1.2.0  
 Id: 10/11

